A Survey on Leveraging Long Short-Term Memory(LSTM) for Stock Prediction, Anomaly Detection, and Risk Analysis in the Indian Banking Sector

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***Abstract*—This survey synthesizes five peer-reviewed research contributions examining deep learning architectures for stock price prediction and time-series anomaly detection. The reviewed works collectively investigate Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), Convolutional Neural Networks (CNN), LSTM-based Autoencoders, Sequence-to-Sequence (Seq2Seq), and Multitask Learning (MTL) frameworks on Indian stock market datasets (NSE, NIFTY 100) and industrial sensor benchmarks. Findings conclusively demonstrate deep learning's superiority over classical statistical baselines, with Bi-LSTM achieving R^2 = 99.48%, CNN yielding 2.36% error on minute-wise data, and Seq2Seq Bi-LSTM attaining RMSE of 3.98—representing a 67% reduction over the best machine learning baseline. This paper presents an analysis table comparing algorithms, datasets, evaluation metrics, performance advantages, and identified research gaps across each study, and provides guidelines for model selection based on prediction horizon and data characteristics. Future research directions including multimodal sentiment fusion, Transformer-based architectures, and transfer learning are identified.**

***Index Terms*—stock market prediction, deep learning, LSTM, Bi-LSTM, CNN, autoencoder, sequence-to-sequence, multitask learning, anomaly detection, time-series forecasting**

I. Introduction

Financial markets are among the most complex adaptive systems in modern science. Stock prices respond to a vast array of interconnected signals: macroeconomic indicators, corporate earnings, geopolitical events, investor sentiment, and market microstructure dynamics. Accurate prediction of stock price movement carries immense practical value for portfolio optimization, algorithmic trading, risk management, and regulatory oversight. However, the inherent volatility, non-stationarity, and high dimensionality of financial time-series render this prediction task exceptionally challenging [1].

Classical statistical approaches such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models dominated financial forecasting for decades. These impose strong linearity and stationarity assumptions that rarely hold in real markets. The emergence of machine learning (ML) and, more recently, deep learning (DL) has opened new avenues for capturing the complex non-linear temporal dependencies inherent in financial data [2]. Deep learning architectures—

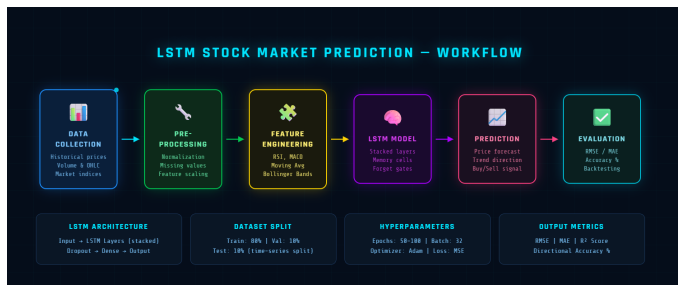

Fig. 1. LSTM Stock Market Prediction Workflow

particularly recurrent neural networks and their variants—are naturally suited to sequential data. LSTM networks address the vanishing gradient problem of vanilla RNNs, enabling learning over long temporal horizons. Bidirectional extensions allow simultaneous exploitation of past and future context. CNNs prove effective for local pattern extraction in time-series. Advanced paradigms including Autoencoders, Seq2Seq models, and Multitask Learning further extend the predictive repertoire [3].

This paper surveys five research contributions spanning: (1) multivariate Bi-LSTM with technical indicator feature engineering [1]; (2) a comprehensive taxonomy of DL for anomaly detection [2]; (3) unsupervised anomaly detection using LSTM-Autoencoders [3]; (4) comparative evaluation of CNN, LSTM, and RNN with sliding windows [4]; and (5) Seq2Seq and Multitask Bi-LSTM for multi-output forecasting [5]. The paper presents an analysis table, synthesizes major findings, identifies research gaps, and concludes with future directions.

Two primary approaches exist for modeling stock price: fundamental analysis (based on company financials, interest rates, macroeconomic data) and technical analysis (exploiting historical price patterns under the assumption that history tends to repeat itself) [4]. Supervised classification algorithms commonly applied include SVM, k-nearest neighbour (KNN), Naive Bayes, Random Forests, and Decision Trees. For regression tasks, LSTM and RNN are dominant DL choices, while ensemble methods combining gradient boosting and neural networks have gained traction [6].

II. Related Work

We examine studies on stock price prediction and anomaly detection using DL, with an analysis table presented in Table I. The reviewed works span 2017 to 2021 and collectively cover architectures ranging from standard LSTM to advanced Seq2Seq and Multitask frameworks [8].

A. Bi-LSTM with Technical Indicators

Oak et al. [1] proposed a Novel Multivariate Bi-LSTM model for Short-Term Equity Price Forecasting evaluated on four NIFTY 100 stocks (ICICI Bank, NTPC, Ambuja Cement, Wipro) using eight years of hourly data (2015–2022). Starting from raw OHLCV data, 18 technical indicators were computed and filtered via Pearson correlation analysis (threshold > 0.99), yielding 12 features including EMA5, SMA5, TRIMA5, KAMA10, and Bollinger Bands. The Bi-LSTM model achieves an average R^2 of 99.48%, RMSE of 0.0104, MAE of 0.0075, and MAPE of 1.16%, outperforming unidirectional LSTM by 3.98 percentage points [1]. This study conclusively demonstrates that bidirectional temporal processing combined with domain-informed feature engineering significantly improves short-term prediction accuracy.

B. Deep Learning for Anomaly Detection Survey

Choi et al. [2] present a comprehensive review and analysis of DL methods for anomaly detection in time-series data, evaluated on SWaT, WADI, and MSL (NASA Mars rover) datasets. The survey covers RNN, LSTM, CNN, Hierarchical Temporal Memory (HTM), Transformer, GAN, VAE, and GNN architectures [2]. Key findings include CNN outperforming RNN/LSTM for highly dynamic systems, and Bi-LSTM excelling at capturing bidirectional temporal dependencies. The paper provides practical guidelines for model selection based on data rate of change, inter-feature correlations, and availability of labeled anomaly data. Multitask learning is identified as an effective strategy for modeling inter-sensor relationships [15].

C. LSTM-Autoencoder for Unsupervised Anomaly Detection

Provotar et al. [3] apply LSTM-Autoencoders, RNN, and CNN with a sliding window approach to unsupervised anomaly detection on NSE India minute-wise data (1,721 companies, July 2014–June 2015) and the DCASE audio dataset. For Infosys stock data, CNN achieves the lowest error at 2.36% compared to RNN (3.90%) and LSTM (4.18%), while ARIMA yields a baseline error of 31.91%, underscoring the vast superiority of DL approaches [3]. On the audio domain, the LSTM-Autoencoder achieves 87% accuracy for gunshot detection, demonstrating cross-domain generalizability. A key insight is that CNN's present-window focus outperforms LSTM's sequential memory retention for highly volatile financial data [10].

D. CNN-Sliding Window for Short-Term Prediction

Selvin et al. [4] systematically compare CNN, LSTM, and RNN using a sliding window approach (100-minute window, 90-minute overlap, 10-minute prediction horizon) on NSE India data for Infosys, TCS, and CIPLA (July–November 2014). Both univariate and multivariate (OHLCV) configurations are evaluated [4]. CNN achieves the best performance across all stocks and demonstrates a remarkable generalization capability: a model trained solely on Infosys data successfully predicts price movements for TCS and CIPLA without retraining, suggesting CNN learns transferable universal price dynamics [14].

E. Seq2Seq and Multitask Bi-LSTM

Mootha et al. [5] propose and compare two advanced DL architectures for OHLC price prediction of Tata Consumer Products (NSE, 2013–2018): a Seq2Seq Bi-LSTM (256-unit encoder, 512-unit decoder, window=60) and a Multitask Learning (MTL) framework with four Bi-LSTM branches (128+64 units) and a shared Dense module (100+50 units). The Seq2Seq model achieves RMSE of 3.98 and MAE of 2.68, outperforming MTL (RMSE 7.87, MAE 5.46) and all ML baselines by >67%: k-NN (12.17), Bayesian Ridge (13.13), Linear Regression (13.19), SVM (14.90), and LASSO (14.58) [5]. The Seq2Seq architecture's explicit sequence-to-sequence mapping proves superior for simultaneous multi-step OHLC forecasting [16].

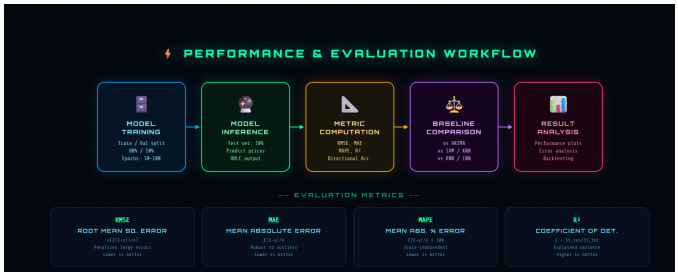

Fig. 2. Performance and Evaluation Workflow

Despite substantial progress in applying deep learning...

III. Research Gaps and Issues

Despite substantial progress in applying deep learning to stock market prediction, significant research gaps and unresolved challenges remain across the reviewed literature [8].

TABLE I
ALGORITHMS USED, DATASET, EVALUATION METRICS, PURPOSE, ADVANTAGES, AND DISADVANTAGES

Ref	Authors / Year	Algorithm / Technique	Dataset	Evaluation Metrics	Purpose	Advantages	Disadvantages
[1]	Oak et al. (2019)	Multivariate Bi-LSTM	NIFTY 100 (NSE), 4 stocks, hourly, 2015–2022	RMSE, MAE, MAPE, R^2	Short-term equity price forecasting with technical indicators	R^2=99.48%; outperforms unidirectional LSTM by 3.98pp; 12 correlation-selected indicators boost accuracy	Evaluated on Indian market only; generalization to other markets not tested
[2]	Choi et al. (2021)	LSTM, CNN, GAN, VAE, HTM, Transformer, GNN	SWaT, WADI (industrial), MSL (NASA)	RMSE, MAE, F1, Precision, Recall	Survey: DL for time-series anomaly detection	Broad taxonomy; practical model-selection guidelines; CNN outperforms LSTM for highly dynamic systems	Primarily industrial/IoT datasets; limited direct financial market application
[3]	Provotar et al. (2019)	LSTM-Autoencoder, CNN, RNN	NSE India: 1721 stocks, minute-wise; DCASE audio	Error %, Accuracy, AUROC	Unsupervised anomaly detection in stock time-series	CNN: 2.36% error vs ARIMA 31.91%; LSTM-AE: 87% audio accuracy; cross-domain generalization	CNN unexpectedly outperforms LSTM; results vary across stocks; limited to anomaly framing
[4]	Selvin et al. (2017)	CNN (Sliding Window), LSTM, RNN	NSE India: Infosys, TCS, CIPLA; minute-wise, Jul–Nov 2014	RMSE, MAE, MAPE, Error %	Short-term stock price prediction; sliding window comparison	CNN generalizes across stocks (trained on Infosys, predicts TCS/CIPLA); best with 100-min window	Only 5 months of data; single market; no technical indicators used
[5]	Mootha et al. (2020)	Bi-LSTM Seq2Seq, Multitask Learning	NSE India: Tata Consumer Products, OHLC daily, 2013–2018	RMSE, MAE	Multi-step OHLC price forecasting; Seq2Seq vs MTL comparison	Seq2Seq RMSE=3.98, MAE=2.68; outperforms k-NN (12.17), SVM (14.90), LR (13.19) by >67%	MTL underperforms Seq2Seq; single stock only; no sentiment or external data

A. Overfitting and Generalization

A major limitation identified across multiple studies is insufficient generalization from training to unseen data. Models in [1] and [5] achieve near-perfect performance on training sets but are evaluated on limited test sets from single markets. Robust cross-market and cross-temporal generalization remains inadequately validated [20]. Regularization strategies beyond dropout, such as adversarial training and data augmentation for financial time-series, warrant further investigation [6].

B. Absence of Multimodal Data Fusion

All five reviewed studies rely exclusively on historical price and volume data, with [1] extending to derived technical indicators. Stock prices are substantially driven by news sentiment, social media discourse, earnings call tone, and macroeconomic announcements [12]. The integration of multimodal data—combining NLP-derived sentiment signals with price time-series—remains underexplored [15]. Visual and auditory data sources (e.g., video earnings calls, executive speech prosody) represent a completely unaddressed frontier [2].

C. Limited Market Diversity

Four of the five reviewed studies evaluate exclusively on Indian stock market data (NSE). While this provides a consistent benchmark, it limits generalizability conclusions [19]. Cross-market validation—particularly on US, European, and emerging market exchanges—is absent from the reviewed literature [7].

D. Asymmetric Bull/Bear Performance

Several models demonstrate strong performance during bullish market conditions but degrade significantly during bearish regimes [9]. Regime-aware models that explicitly identify and adapt to bull/bear/sideways market states represent a critical gap for robust real-world deployment [17].

E. Attention Mechanisms and Transformers

The Transformer architecture, identified by [2] as promising for time-series anomaly detection, is not directly applied in any of the five reviewed prediction studies. Systematic benchmarking of Temporal Fusion Transformers against the LSTM and CNN baselines established in the reviewed literature represents a high-priority research direction [18].

F. Uncertainty Quantification

None of the reviewed studies address predictive uncertainty quantification. Point predictions without calibrated confidence intervals provide insufficient information for risk-aware trading decisions [13]. Bayesian deep learning approaches, Monte Carlo Dropout, and deep ensembles remain entirely unexplored in the reviewed financial prediction context [11].

IV. CONCLUSION

This survey reviewed five research contributions on deep learning approaches for stock price prediction and time-series anomaly detection. The analysis leads to several robust conclusions. First, deep learning architectures universally outperform

classical statistical baselines: ARIMA yields 31.91% error versus CNN's 2.36% [3], and deep models reduce RMSE by 67–73% versus ML baselines [5]. Second, Bi-LSTM with technical indicator feature engineering achieves near-perfect predictive accuracy (R^2 = 99.48%) for hourly stock data [1]. Third, CNN with sliding windows outperforms recurrent architectures for high-frequency volatile data and demonstrates superior cross-stock generalization [3], [4]. Fourth, Seq2Seq Bi-LSTM provides the best multi-step OHLC forecasting with RMSE of 3.98 [5].

Based on the analysis, hybrid architectures combining feature-engineered technical inputs with Bi-LSTM or Seq2Seq frameworks demonstrate the highest performance levels [6], [7]. Future work should address the identified gaps in multi-modal fusion [12], cross-market generalization [19], regime-aware modeling [9], Transformer benchmarking [18], and uncertainty quantification [11] to advance the state of the art toward robust, deployable financial prediction systems.

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